

Doc. 267 – Cosmological Degeneracy: Λ CDM and FFGFT

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Correction notes

Corrections from the calculation review (1 Oct. 2026). The following passages are corrected or annotated in the text; each carries a dated note with the previous value:

- Note: the factor 3 in front of $\{1, 6, 14, 26\}$ does not follow from the projection onto three spatial directions; it is a choice (Doc. 190 R139, Doc. 268).

Purpose and Core Thesis

This document compares the two cosmological models – the Λ CDM standard model and FFGFT – in their respective internal logic. The core thesis:

The principal cosmological standard measurements (Type Ia supernovae, baryon acoustic oscillations, CMB temperature) are *degenerate*: they cannot in principle distinguish a physical motion of space (Λ CDM) from an intrinsic, non-linear unfolding of the time metric (FFGFT) – as long as both models produce the same distance-redshift relation $z(d)$.

This is not a claim of FFGFT's superiority. It is a methodological observation: the data alone do not decide between the models. The decision shifts to the level of theoretical consistency, parameter count, and derivability from first principles.

.1 The Two Models in Their Basic Structure

The inverted initial condition

The fundamental difference is not in the predictions but in the direction of evolution:

Phase	Λ CDM (standard model)	FFGFT
Beginning	Singularity, Big Bang, extremely hot and dense, time passes fast	Minimal length $L_0 = \xi \cdot \ell_P$, nearly compactified, concentrated but cold, time passes extremely slowly
Evolution	Metric expansion of space + cooling	Fractal unfolding of time $\rightarrow$ apparent expansion
Today	Expanding, cold, accelerated by dark energy (Λ)	Apparently expanding, stationary quantum ground state, apparently accelerated by non-linear time recursion

The two directions are fully inverted:

- Λ CDM: *hot and fast* $\rightarrow$ *cold and accelerated*
- FFGFT: *cold and slow* $\rightarrow$ *apparently warm and accelerated* (through time unfolding)

Temperature in both pictures

In Λ CDM, today's CMB temperature ($\approx 2,7$ K) is the cooled remnant of a hot early phase. In FFGFT it is not a remnant but a stationary equilibrium value from the ξ -field quantisation (Doc. 025): the radiation does not cool, it is an inherent feature of the vacuum in equilibrium. In FFGFT, the initial concentration is *topological* (winding numbers close together), not *thermodynamic* (hot).

.2 The Cosmological Degeneracy – Three Examples

Type Ia Supernovae and Light-Curve Stretching

The signal: The data show $b \approx 1$: a distant supernova runs more slowly for us (light curve stretched), and its light is redshifted.

Λ CDM: Space expands metrically during the photon flight, stretching wavelength (red-shift) and pulling apart signal duration.

FFGFT: No spatial expansion. But the fractal time unfolding lengthens the path the signal travels through the geometry. This geometric path lengthening stretches wavelength and pulse duration in the exact same ratio – hence also $b = 1$.

The indistinguishability: Both mechanisms yield exactly the same mathematical curve. The DES value $b = 1,003 \pm 0,011$ excludes a *non-dilating* static model ($b = 0$) at $\sim 90\sigma$ – but FFGFT predicts $b = 1$, not $b = 0$. FFGFT is therefore not affected by this test.

Baryon Acoustic Oscillations and Angular Scales

The signal: One measures the angular separation $\theta(z)$ of galaxy clusters.

Λ CDM: The apparent angle changes because the scale factor $a(t)$ grows and dark energy accelerates the expansion.

FFGFT: The apparent acceleration results from the non-linear temporal recursion. Since the effective time rate per distance step changes, the $z(d)$ relation curves exactly as if there were accelerated expansion.

The indistinguishability: As long as the FFGFT time evolution produces the same $z(d)$ relation, it yields the same angles on the sky. The BAO data contain no marker for which mechanism produced the curve.

Important caveat (P20): The BAO test uses D_M/r_s , where $r_s \approx 147$ Mpc was computed under Λ CDM assumptions. Onur's $\Delta\chi^2 \approx 138$ compares a logarithmic distance law against Λ CDM pipeline data – model against model, not theory against raw nature.

The Cosmic Microwave Background and the Acoustic Peaks

The signal: A nearly perfect blackbody spectrum at $\approx 2,7$ K with acoustic peaks at $\ell \approx 200, 500, 800, \dots$

Λ CDM: Sound waves in the photon-baryon plasma, frozen at recombination. The first peak ($\ell \approx 200$) marks regions at maximal compression; higher peaks are overtones.

FFGFT approach: No hot phase, no plasma. The CMB temperature is the stationary quantum ground state of the Φ -field (Doc. 025). The peaks would arise from discrete T^4 lattice resonances – like the emission lines of an atom or phonon dispersion in a crystal.

The ratio $(\lambda_e/L_0)^{1/\ln(1/\xi)}$ with the reduced Compton wavelength λ_e and $L_0 = \xi \cdot \ell_p$ gives numerically:

$$\left(\frac{\lambda_e}{L_0}\right)^{1/\ln(1/\xi)} = \left(\frac{3,86 \times 10^{-13}}{2,16 \times 10^{-39}}\right)^{1/8,92} \approx 875.$$

This value lies in the correct order of magnitude, close to Λ CDM recombination ($z \approx 1089$).

Caveat (P20): The exponent $1/\ln(1/\xi)$ in the power is not derived from the FFGFT documents available; its justification from the T^4 geometry remains open. z_* is therefore currently a *reference value* of the right order of magnitude, not a derived FFGFT result.

.3 The Peak Structure from Pure Geometry

Acoustic resonators are exactly computable, and their eigenmode ratios depend *only on the geometry* of the resonator, not on the medium: a pipe or sphere (j_0) gives $1 : 2 : 3 : 4 : 5$, a half-open pipe $1 : 3 : 5 : 7$, a circular membrane (J_0) $1 : 2,30 : 3,60$. The CMB ($1 : 2,44 : 3,68$) is an acoustic series with an offset.

The key point: the peak structure is set by the *geometry* of the resonance space, not the medium. Λ CDM: sound horizon in plasma; FFGFT: T^4 torus geometry. Both can produce the same series.

Concrete T^4 calculation. The thermally weighted spectrum $I(r) = g(r) \cdot \langle N(r) \rangle \cdot r$ with Bose-Einstein occupation $\langle N \rangle = 1/(e^r - 1)$ and Jacobi degeneracy $g(r)$ has local maxima at $|n|^2 = 3, 6, 10, 14, 18, \dots$. The ratios $\ell \propto \sqrt{|n|^2}$ of the *dominant* peaks $|n|^2 = 3, 18, 42, 78$ (all genuine local maxima) match the CMB series:

$$\sqrt{3} : \sqrt{18} : \sqrt{42} : \sqrt{78} = 1 : 2,449 : 3,742 : 5,099 \quad \text{vs.} \quad 1 : 2,441 : 3,682 : 5,091 \text{ (CMB)}.$$

Deviation $< 2\%$ with no free parameter.

The factor 3 counts the three observable spatial dimensions. (Note of 1 Oct. 2026: A spatial projection cannot produce the factor 3, since $n_1^2 + n_2^2 + n_3^2 \leq |n|^2$; for $(1, 1, 1, 0)$ one already has $|n|^2 = 3$, so this mode does not belong to $|n|^2 = 1$. $\{3, 18, 42, 78\} = 3 \cdot \{1, 6, 14, 26\}$ is a choice, not a consequence of projection (Doc. 190 R139; note in Doc. 268, Step 8). The ratios $\sqrt{3} : \sqrt{18} : \sqrt{42} : \sqrt{78}$ do not depend on the factor.) The peaks 3, 18, 42, 78 are not arbitrary – they are exactly $3 \cdot \{1, 6, 14, 26\}$. The factor 3 arises because the observer measures

in 3D space (plus time-unfolding): a T^4 mode $\vec{n} = (n_1, n_2, n_3, n_4)$ is perceived not through its full torus radius $|n|^2$ but through the spatial projection

$$k_{\text{obs}}^2 = \frac{n_1^2 + n_2^2 + n_3^2}{L_{\text{tor}}^2}, \quad \ell \propto k_{\text{obs}}.$$

For the most symmetric ground mode $(1, 1, 1, 0)$ – all three spatial directions equal, $n_4 = 0$ – one has $n_1^2 + n_2^2 + n_3^2 = 3$. This shifts the T^4 ground peaks $|n|^2 = 1, 6, 14, 26$ to the observable series 3, 18, 42, 78. The fractal spatial dimension $D_f = 3 - \xi \approx 3$ is *consistent* with this (the factor is the spatial dimension, slightly modified by ξ , correction $\xi/3 \approx 4.4 \times 10^{-5}$), but the origin of the factor is the 3D projection, not a dispersion rescaling. The detailed derivation and the honest caveat (the rigorous selection mechanism via the spherical C_ℓ projection remains open, P30) are in Doc. 268, Steps 8–9.

Two filters produce the peak structure. The observed discrete, sharp CMB peaks follow from two properties of the same complete T^4 acting together – seen from two reference points:

- **Filter 1 – compactness (the T^4 as a whole, from outside):** A compact (finite) space gives a *discrete* mode spectrum; an infinite-open space gives a continuous one – no sharp peaks. This filter forces the *existence* of discrete peaks.
- **Filter 2 – fractal structure (from inside, decompactified):** The fractal granulation ($D_f = 3 - \xi$, Jacobi degeneracy of the four-square sum and Bose-Einstein weighting) weights the discrete modes unequally. Only this produces the *dominant* peaks at $|n|^2 = 3, 6, 10, 14, \dots$ rather than a uniform lattice. This filter forces the *location* of the peaks.

The two filters are not two different objects but two reference points on the same complete T^4 : compact-finite from outside (Filter 1, discrete modes), fractally granulated inside down to $L_0 = \xi \cdot \ell_p$ (Filter 2). The fractal non-closure is therefore not a *second* property contradicting compactness, but the inside view of the same closed torus – a matter of reference point (inside/outside, P20). Only both filters together give the observed CMB series; the discrete peaks thus argue *for* the compact T^4 and *against* a purely-infinite-open universe.

Honest caveat: This conclusion is FFGFT-internal. The peaks do *not* force spatial compactness – Λ CDM produces the same discrete peaks from a *temporal* bound (the sound horizon at recombination) in a spatially infinite universe. This is the cosmological degeneracy: the peaks alone do not distinguish spatial compactness (FFGFT) from a temporal bound (Λ CDM).

.4 Peak Ratios and Absolute Scale

Ratios: no free parameter

The peak ratios follow directly from the T^4 geometry – no fit, no external parameter:

$$\sqrt{3} : \sqrt{18} : \sqrt{42} : \sqrt{78} = 1 : 2,449 : 3,742 : 5,099$$

Observed: 1 : 2,441 : 3,682 : 5,091. Deviation < 2%.

Caveat: if peak positions come from a Λ CDM template fit, Λ CDM is already embedded. A clean test needs model-free local maxima of C_ℓ (~ 1 –2% deviation from Λ CDM-fit values).

Absolute scale: R_H from ξ

The absolute position $\ell_1 \approx 220$ requires a length scale. In FFGFT this is R_H – and R_H is fully derived from ξ (Doc. 182):

$$\xi \rightarrow E_0 = \sqrt{m_e m_\mu} \rightarrow E_H = E_0 \cdot \xi^{41/4} \rightarrow R_H = \frac{\hbar c}{E_H} \approx 1,40 \times 10^{26} \text{ m}$$

No free parameter. R_H is simultaneously the Schwarzschild radius of the total mass $M_U = R_H c^2 / (2G)$ of the static universe – an internal consistency check (Doc. 182, Doc. 190 P4).

Quantity	Origin	Status
Ratios 1 : 2,449 : ...	T ⁴ geometry, Jacobi degeneracy	No free parameter
R_H (absolute scale)	$\xi \rightarrow E_H \rightarrow R_H$ (Doc. 182)	No free parameter
Link $R_H \rightarrow \ell_1$	Effective resonator radius	Open (P20)

Script: ffgft_lambda_im_messprozess.py.

.5 Inversion and the Symmetric Circularity

Starting from the measured peaks, the fundamental angle is $\theta_1 = \pi/A \approx 0,59^\circ$. But $\theta_1 = L_{\text{res}}/D_A$ is a *ratio*: only the product $H_0 \cdot L_{\text{res}} = \theta_1 c / \ln(1 + z_*) \approx 458 \text{ km/s}$ is determined, not H_0 alone. Isolating H_0 requires fixing one length independently – the external parameter of P20.

This makes the circularity **symmetric**:

	ΛCDM	FFGFT
Measured	θ_1 (angle)	θ_1 (angle)
External length scale	$r_s \approx 147 \text{ Mpc}$	R_H (torus size)
Source of scale	hot early phase (model)	$E_H/\hbar$ from ξ
H_0	67,4 (output)	66,2 (set)
Circular?	Yes	Yes

A measured angle always determines only a length ratio. Both models must insert an absolute scale. No one obtains a model-neutral H_0 from the CMB peaks. The fair comparison is not “who is circularity-free” (neither is) but parsimony (FFGFT: one external scale; ΛCDM : r_s from 6 parameters plus the BBN phase) versus anchoring (ΛCDM better tested; FFGFT’s early phase open).

Scripts: ffgft_zirkularitaet_symmetrie.py, lcdm_circularity.py.

.6 Accelerated Dilation in Both Models

An often overlooked point: both models predict an *accelerated* dilation, not a constant one.

Aspect	Λ CDM	FFGFT
Cause	Dark energy (Λ)	Fractal temporal evolution ($\mathcal{R}$)
Mechanism	Repulsive gravitation	Accumulated path lengthening
w	≈ -1	No w (no Λ field)
Acceleration	Real (metric expansion)	Apparent (emergent time dilation)

In FFGFT, the effective scale change per time step grows with the number of recursion steps: the farther we look in distance (cosmologically: into the past), the more unfolded time lies between events. The observation of an accelerated dilation is therefore no contradiction to FFGFT – it is exactly what FFGFT expects from its own dynamics. Only the explanation differs.

This further complicates the distinction: many observations (SNe Ia, BAO) primarily measure the acceleration itself, not its mechanism.

The three phases of the Λ CDM expansion history

A structural difference concerns the *temporal course* of the expansion rate. Λ CDM requires three successive regimes:

1. **Early phase (inflation):** an extremely rapid, exponentially accelerated expansion driven by a hypothetical inflaton field with negative pressure, which then decays.
2. **Intermediate phase (decelerated expansion):** radiation- and matter-dominated, $a(t) \sim t^{1/2}$ resp. $t^{2/3}$. Attractive gravitation *slows* the expansion.
3. **Present phase (renewed acceleration):** from $z \approx 0,6$ the cosmological constant Λ dominates and drives the expansion in an accelerated manner again.

The sequence *accelerated* $\rightarrow$ *decelerated* $\rightarrow$ *accelerated again* is mathematically allowed by the Friedmann equation, but it requires *two separate, independently postulated* acceleration mechanisms (inflaton and Λ), separated by a braking phase. There is also the coincidence problem: Λ becomes dominant precisely in our cosmic epoch – a fine-tuning the model does not explain from within.

FFGFT, by contrast, manages with *one* mechanism: the apparent acceleration grows monotonically with the recursion depth of the fractal time unfolding; there is no separate inflationary and no separate Λ epoch. This is an advantage at the level of theoretical economy, *not* an empirical statement – FFGFT has no quantitative predictions for the early phase so far, and whether its single unfolding dynamics reproduces the entire expansion history correctly has not been shown.

.7 The Limit for Both Models: No Model-Neutral Scale

The degeneracy is reinforced by a second problem affecting both models equally (P20, Doc. 190): there is no model-neutral determination of the cosmological scale R_H .

- Λ CDM measures $H_0 \approx 67,4$ km/s/Mpc – but this value is a pipeline output presupposing expansion, the FLRW metric, and the matter content.
- FFGFT derives $E_H/\hbar \approx 66,2$ km/s/Mpc internally from $\xi^{41/4}$ – and thus also $R_H = \hbar c/E_H$ fully from ξ (Doc. 182). What remains open is the precise link $R_H \rightarrow \ell_1$ (how R_H yields the absolute peak position, P20).

Onur's tests use Λ CDM pipeline quantities (r_s , H_0 , calibrated magnitudes) as reference. They therefore test not "static vs. expanding" but " Λ CDM pipeline vs. the absence of this pipeline." That is not a neutral test.

.8 Where the Decision Actually Falls

If the data are blind to the mechanism, the decision shifts to the theoretical level:

1. **Parameter count:** FFGFT needs internally only the fundamental ratio $\xi = 4/30000$. From this $R_H = \hbar c/E_H$ follows fully (Doc. 182); what remains open is the precise link $R_H \rightarrow \ell_1$ (P20). Λ CDM requires parameters for dark energy, dark matter, inflation, baryon density, and more.
2. **Singularities:** FFGFT produces no singularity (no Big Bang, no infinite density at $t = 0$). Λ CDM begins with a singularity.
3. **Explanation of time:** FFGFT derives the emergence and change of time flow (the fractal unfolding) from first principles. Λ CDM posits time as a rigid, unexplained background stage.

Important – no superiority claim: These points are advantages at the level of theoretical economy and consistency. They are *not* an empirical refutation of Λ CDM.

.9 The Honest Scientific Position

- There are two competing explanations for the same observations.
- Neither is currently empirically preferred, because the classical measurements (SNe Ia, BAO, CMB temperature) are degenerate.
- The data do not force a choice between the models.
- The choice is a question of theoretical consistency and explanatory power beyond the cosmological standard data.
- FFGFT is neither confirmed nor refuted by Onur's tests. The tests show only that a *naïve* static model without fractal time unfolding ($b = 0$, no acceleration) is excluded. FFGFT is not such a naïve model.

The point where the models could in principle differ: The early phase. Caution is needed here, however – the early phase is hardly directly verifiable for *either* model, and likely will remain so: no one can directly observe the first moments of the universe; both models reconstruct it from present-day traces and many assumptions. Λ CDM has had decades to develop a detailed explanatory framework for this phase (hot Big Bang, inflation, nucleosynthesis) – successful in fitting the data, but itself built on numerous assumptions and free parameters. FFGFT would have to explain the light-element abundances (BBN), the CMB acoustic peaks, and structure formation from its inverted direction (cold, slow, fractally unfolding) – an open research question on which little work has been done.

The early phase is thus in principle the place where the models could diverge – but given the limited verifiability for both sides, it is not to be expected that it will lead to a clear empirical decision. A quantitative FFGFT prediction for the early observables would be desirable but is still outstanding.

Related documents: P18–P20 (Doc. 190, status of the cosmological scale), Doc. 025 (CMB as ξ -field ground state), Doc. 182 (fractal path lengthening), Doc. 264 (ferrotoroidicity, $P''=0$ as static ground state), Doc. 266 (analysis of Onur's tests).